PROJECT DEVELOPMENT PHASE

Project Title: ComicCraft –AI Comic Story Creator using Gemini Models

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1. Introduction

The development phase of ComicCraft focuses on turning the project plan into a working application. The system uses Google Gemini models to help users create comic stories from simple text prompts. It generates a story outline, divides it into panels, and provides character descriptions, dialogue, captions, and illustration prompts. The application is designed to make comic creation easier for students and beginners.

2. Development Objectives

- Build a simple and user-friendly interface for entering comic ideas.
- Connect the application to an available Gemini model through an approved API or supported interface.
- Generate a coherent story based on the user's prompt.
- Organize the story into a selected number of comic panels.
- Generate dialogue, captions, character descriptions, and illustration prompts.
- Allow users to review, edit, regenerate, and copy the generated content.

3. System Architecture

ComicCraft follows a simple three-layer architecture. The presentation layer collects input and displays results. The application layer validates input, builds prompts, sends requests, and formats responses. The AI service layer uses the Gemini model to

generate story content. If a backend is used, it securely handles the API request so that the API key is not exposed in browser code.

Presentation Layer — User Interface (topic, genre, panel count, results)
Application Layer — Input validation, prompt building, API handling, response formatting
AI Service Layer — Gemini model generates story and panel content

4. Tools and Technologies

Tool / Technology	Purpose
Google Gemini model/API	AI-assisted story and panel generation
HTML and CSS	Page structure and visual design
JavaScript	User interaction and application logic
Python or Node.js (optional backend)	Secure API calls and request processing
Web browser	Run and test the application
Code editor	Write and maintain source code

5. Implementation Steps

1. Create the project folder and prepare the development environment.
2. Design the user interface with fields for topic, genre, audience, and number of panels.
3. Add input validation to ensure required fields are completed.
4. Create a prompt template that asks Gemini for a title, characters, story outline, and panel-by-panel content.
5. Connect the application to the selected Gemini model using the supported API or interface.
6. Process the model response and display each panel in a readable format.
7. Add options to edit, regenerate, and copy the generated comic plan.
8. Handle errors such as network failure, invalid input, or API limits.

6. Prompt Design

A structured prompt helps the model produce a consistent result. The prompt should specify the topic, genre, target audience, number of panels, character details, and

output format. It should ask for a clear beginning, middle, and ending, with concise dialogue and a visual description for every panel.

Example prompt: "Create a 5-panel comic about a student who discovers a mysterious seed. Use a friendly adventure style for school students. First provide a title and character descriptions. For each panel, include the scene, action, dialogue or caption, and an illustration prompt. Keep the story coherent and end with a positive message."

7. Sample Output Structure

Panel	Scene / Action	Dialogue / Caption	Illustration Prompt
1	Introduce the main character and setting.	"Today will be an adventure!"	A student in a school garden.
2	The character finds a glowing seed.	"What is this?"	A tiny glowing seed in the soil.
3	The seed begins to grow.	"It's magical!"	A small plant growing with sparkling leaves.
4	Friends help care for the plant.	"Let's protect it together."	Students watering a special plant.
5	The plant blooms and brightens the garden.	"Teamwork makes things grow!"	A colourful garden with happy students.

8. Core Functionalities

- Prompt entry: users enter a story idea.
- Genre and panel selection: users customize the comic.
- AI story generation: Gemini creates the story content.
- Panel formatting: content is organized panel by panel.
- Editing and regeneration: users can improve the output.
- Copy/export: users can save the comic script for later use.

9. Testing During Development

Test Case	Expected Result	Status
Valid topic submitted	A comic story is generated.	To be tested
Empty topic submitted	A validation message is shown.	To be tested
Different panel counts	Output follows the selected count.	To be tested
API/network error	A helpful error message appears.	To be tested
Edit or copy output	User can modify or copy content.	To be tested

10. Security and Responsible Use

- Keep API keys in secure server-side configuration; do not publish them in frontend code.
- Validate user input and avoid sending unnecessary personal information.
- Review generated stories for factual errors, bias, and age-appropriateness.
- Respect copyright and avoid asking the system to reproduce existing comic scripts or protected characters inappropriately.
- Tell users that AI-generated content may need human review.

11. Challenges and Solutions

Challenge	Planned Solution
Story details may change between panels	Include a shared character and setting description in the prompt.
Output may not follow the requested format	Use clear formatting instructions and validate the response.
Slow response or service interruption	Show progress, handle timeouts, and provide a retry option.
API usage limits	Monitor usage and provide clear error handling.
Inconsistent image style	Use a consistent visual-style description in each illustration prompt.

12. Development Deliverables

- User interface for comic story creation.
- Gemini integration or a working mock interface if API access is not available.
- Prompt templates and response formatting logic.
- Panel-by-panel comic storyboard output.
- Basic validation and error handling.
- Test results and project documentation.

13. Conclusion

The development phase transforms the ComicCraft plan into an interactive comic-story creation prototype. By combining a simple interface, structured prompts, and Gemini-powered text generation, the application can help users turn ideas into organized comic panels. After implementation, the prototype will be tested and refined before final demonstration.